The Table TELCO_CHURN1 has over 1 million Data Rows and 21 columns as follows :

Churn : Value is YES if Customer is churned and NO if Customer is not Churned.

Contract : Contract types of Customers are of Three types –Month-To-Month,One Year,Two Year

customerID : Unique Customer identity Number

Dependents : Number of Dependent of Customers (Yes/No)

DeviceProtection : Whether the Customers has subscribed to Device Protection (Yes/No)

gender : Customers gender (Male/Female)

InternetService : Whether the Customers has subscribed to Internet Service(Yes/No)

MonthlyCharges : Monthly charges of Customers.

MultipleLines: Whether the Customers has Multiple lines(Yes/No)

OnlineBackup : Whether the Customers has subscribed to Online Backup Service (Yes/No)

OnlineSecurity: Whether the Customers has subscribed to Online Security Service(Yes/No)

PaperlessBilling: Whether the Customers has subscribed to Paperless Billing(Yes/No)

Partner : Whether the Customers has a partner (Yes/No)

PaymentMethod : The type of Payment Method used by Customers viz:Bank Transfer,Credit Card,Electronic check,mailed Check

PhoneService: Whether the Customers has Phone Service(Yes/No)

SeniorCitizen : Whether the Customers is Senior Citizen(Yes/No)

Streaming_movies : Whether the Customers has subscribed to Streaming movies service (Yes/No)

StreamingTV : Whether the Customers has subscribed to Streaming TV Service(Yes/No)

TechSupport : Whether the Customers has availed tech Support service (Yes/No)

Tenure : Tenure period of Customers in months

TotalCharges : Total Charges of Customers till Date.

Data Cleaning in Power BI Query :

We are doing Data Cleaning in Power BI through “Transform Data “

Step I: Fixing Data Types of all Columns wherever required.

The Columns " Monthly Charges", and " Total Charges " changed from TEXT to DECIMAL Data types by selecting the column and further selecting "Change type ": option upon right click.

The Data type of column "TENURE" changed from TEXT to "WHOLE NUMBER" .

Likewise the data type of all other TEXT columns should be checked for TEXT type.

CHURN Column-→ TEXT Data type.

Step II: Column TOTAL_CHARGES

Fixing NULL and BLANK Values in TOTALCHARGES Column by changing to ZERO as the Customers seems NEW with NOT NULL Values in MONTHLY_CHARGES Column.

Step III : Column MONTHLY_CHARGES

Removing $ sign before values by Replacing "$" values with null.

Step IV: **Column "CONTRACT" Standardization.**

The Column "CONTRACT" has multiple values for Contract like " Month to Month " and "Month-To-Month". The column values are changed to only one type of Values "Month-To-Month" by right clicking the Column and choosing "Replace Values "

Step V : **Capitalize Each Word of Text Columns.**

The Columns whose data types are of TEXT types need common understanding for proper utilization by capitalizing first letter of each word of text columns.

Step VI : Removing Extra spaces in Row Values.

The Row values in table may have extra spaces before and after the Values. This needs to be fixed as it may interfere in data calculations. The Extra spaces are removed by "TRIM " feature available in "FORMAT" menu . All such text columns are selected first using CTRL key and then the TRIM feature is used. This will remove all extra spaces in all values of Table. More Specifically on TEXT columns values.

Step VII : **Data Cleaning by "CLEAN" feature .**

The Data cleaning is also done by using "CLEAN" option available in FORMAT in TRANSFORM DATA.

This will automatically fix the error in data values like assigning unique values to Data values.

Step VIII: Fixing Logical Errors

The Logical error needs to be checked for Data Accuracy and Cleaning. With Industry and domain knowledge , the logic is used to interpret errors in data and make it correct.

The Column PHONE_SERVICE has values YES/NO. If the Customer do not have Phone Service, He//She cannot have Multiple Lines or Internet Service or Streaming Movies or Streaming TV.

Thus the data values should be NO for columns such as MULTIPLE_LINES or INTERNET_SERVICE.

Case 1:

Phone Service=No

Multiple lines= Yes 

Solution :

If Phone Service=No

Multiple Lines=No 

The Solution to fix this logical error is by using “Conditional Column” feature in “Add Column “ menu in TRANSFORM DATA option.

Condition :

- If Phone Service =No -→ Multiple lines=No
- Else keep existing values

Steps to do :

Use Conditional Column as Follows:

New Colum name : MULTIPLE_LINES2

Column Name: Select “PHONE_SERVICE”

Operator: ”EQUAL”

VALUE :No

Then Output: No

Else :(Select Column “ MULTIPLE_LINES”

This logic will create a new column named MULTIPLE_LINES2. The Column values will depend on values of “PHONE SERVICE” Column. If the PHONE SERVICE Value is NO, Then the MULTIPLE_LINES value is also NO.

ELSE the MULTIPLE_LINES values will be same.

Case 2:

- If Internet Service=No Then
- Streaming TV=Yes 

Solution :

- If Internet Service=No Then
- Streaming TV=No 

Likewise the Column INTERNET SERVICE has values YES/NO. if the Customer do have INTERNET SERVICE then(Value-NO) then the Column values in STREAMING_MOVIES or STREAMING TV should be also be NO. This is a data error and it needs to be fixed. The data is accordingly replaced with NO values by creating a new Conditional Column as done above.

Step IX: Removing Duplicate values in CUSTOMER ID column.

The Duplicates values in CUSTOMERID column can be removed by selecting the column and selecting REMOVE DUPLICATES option